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Optical imaging system

Abstract

An optical imaging system includes a plurality of fixed lenses disposed along an optical axis; a first reflective member disposed on an object side of the plurality of lenses; and a plurality of reflective members disposed on an image side of the plurality of lenses. At least one of the plurality of lenses is a variable lens having a variable focal length, and each of the plurality of reflective members is configured to move as the focal length of the variable lens changes.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims the benefit under 35 USC 119(a) of Korean Patent Application No. 10-2021-0022588 filed on Feb. 19, 2021 in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Field

(2) The following description relates to an optical imaging system.

2. Description of Background

(3) Recently, a camera module has been employed in portable electronic devices including smartphones.

(4) Also, to indirectly implement an optical zoom effect, a method of mounting a plurality of camera modules having different focal lengths in a portable electronic device has been suggested.

(5) However, this method may require a plurality of camera modules having different focal lengths for the optical zoom effect, a structure thereof may be complicated.

SUMMARY

(6) This Summary is provided to introduce a selection of concepts in simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(7) Various examples provide an optical imaging system which may implement a zoom function by changing a focal length.

(8) In one general aspect, an optical imaging system includes a plurality of fixed lenses disposed along an optical axis; a first reflective member disposed on an object side of the plurality of lenses; and a plurality of reflective members disposed on an image side of the plurality of lenses. At least one of the plurality of lenses is a variable lens having a variable focal length, and each of the plurality of reflective members is configured to move as the focal length of the variable lens changes.

(9) The variable lens may be configured to have a first focal length or a second focal length as the focal length changes, the first focal length may be positive, and the second focal length may be

negative.

(10) The variable lens may include a liquid lens and a flat lens attached to an image-side surface of the liquid lens.

(11) A radius of curvature of an object-side surface of the liquid lens may be variable and a thickness on an optical axis of the liquid lens may be variable.

(12) The variable lens may be configured to have a first focal length or a second focal length as the focal length changes, the first focal length may be positive, and the second focal length may be negative. When the variable lens has the first focal length, the radius of curvature of the object-side surface of the liquid lens may be positive, when the variable lens has the second focal length, the radius of curvature of the object-side surface of the liquid lens may be negative, and the thickness on the optical axis of the liquid lens may be smaller when the variable lens has the second focal length than when the variable lens has the first focal length.

(13) The plurality of lenses may include a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens disposed in order from the object side, and the variable lens may be the second lens or the sixth lens.

(14) The variable lens may be the second lens, the first lens may have positive refractive power, the second lens may have positive or negative refractive power, the third lens may have negative refractive power, the fourth lens may have positive refractive power, the fifth lens may have negative refractive power, and the sixth lens may have positive refractive power.

(15) The variable lens may be the sixth lens, the first lens may have positive refractive power, the second lens may have negative refractive power, the third lens may have positive refractive power, the fourth lens may have negative refractive power, the fifth lens may have positive refractive power, and the sixth lens may have positive or negative refractive power.

(16) The plurality of reflective members may include a second reflective member and a third reflective member, the second reflective member may include a reflective surface configured to refract light in a direction perpendicular to an optical axis of the plurality of lenses, and the third reflective member may include a reflective surface configured to refract light reflected by the second reflective member in a direction parallel to the optical axis of the plurality of lenses.

(17) The second reflective member and the third reflective member may be configured to move together.

(18) The optical imaging system may satisfy $10 < f_{v_1}/D_{6R_1}$, where f_{v_1} is a first focal length of the variable lens, and D_{6R_1} is a distance on an optical axis from a rearmost lens of the plurality of lenses to a reflective member most adjacent to the rearmost lens among the plurality of reflective members when the variable lens has the first focal length.

(19) The optical imaging system may satisfy $-15 < f_{v_2}/D_{6R_2} < -3$, where f_{v_2} is a second focal length of the variable lens, and D_{6R_2} is a distance on an optical axis from a rearmost lens of the plurality of lenses to a reflective member most adjacent to the rearmost lens among the plurality of reflective members when the variable lens has the second focal length.

(20) The optical imaging system may satisfy $-7 < (f_{v_1} \times D_{6R_2}) / (f_{v_2} \times D_{6R_1}) < -1$, where f_{v_1} is a first focal length of the variable lens, f_{v_2} is a second focal length of the variable lens, D_{6R_1} is a distance on an optical axis from a rearmost lens of the plurality of lenses to a reflective member most adjacent to the rearmost lens among the plurality of reflective members when the variable lens has the first focal length, and D_{6R_2} is a distance on an optical axis from the rearmost lens to the reflective member most adjacent to the rearmost lens when the variable lens has the second focal length.

(21) The optical imaging system may satisfy $-3 < f_{v_2}/f_{v_1} < 0$, where f_{v_1} is a first focal length of the variable lens, f_{v_2} is a second focal length of the variable lens, the first focal length is positive, and the second focal length is negative.

(22) The optical imaging system may satisfy $3 < D_{6R_2}/D_{6R_1} < 6$, where D_{6R_1} is a distance on an optical axis from a rearmost lens of the plurality of lenses to a reflective member most adjacent to

the rearmost lens among the plurality of reflective members when the variable lens has a first focal length, and D6R_2 is a distance on an optical axis from the rearmost lens to the reflective member most adjacent to the rearmost lens when the variable lens has a second focal length, and wherein the first focal length is positive and the second focal length is negative.

(23) The optical imaging system may satisfy $0 < L1/TTL1 < 1$, where L1 is a linear distance in a direction parallel to an optical axis from an object-side surface of the forwardmost lens of the plurality of lenses to an imaging surface when the variable lens has a first focal length, and TTL1 is a distance on the optical axis from the object-side surface of the forwardmost lens to the imaging surface when the variable lens has the first focal length.

(24) The optical imaging system may satisfy $0 < L2/TTL2 < 1$, where L1 is a linear distance in a direction parallel to an optical axis from an object-side surface of the forwardmost lens of the plurality of lenses to an imaging surface when the variable lens has a second focal length, and TTL1 is a distance on the optical axis from the object-side surface of the forwardmost lens to the imaging surface when the variable lens has the second focal length.

(25) The optical imaging system may satisfy $1 < (L1 \times TTL2)/(L2 \times TTL1) < 3$, where L1 is a linear distance in a direction parallel to an optical axis from an object-side surface of the forwardmost lens of the plurality of lenses to an imaging surface when the variable lens has a first focal length, L2 is a linear distance in the direction parallel to the optical axis from the object-side surface of the forwardmost lens to the imaging surface when the variable lens has a second focal length, TTL1 is a distance on the optical axis from the object-side surface of the forwardmost lens to the imaging surface when the variable lens has the first focal length, and TTL2 is a distance on the optical axis from the object-side surface of the forwardmost lens to the imaging surface when the variable lens has the second focal length.

(26) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating a first example in which a first lens group and a second lens group are disposed in a first position in an optical imaging system.

(2) FIG. 2 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 1.

(3) FIG. 3 is a diagram illustrating a first example in which a first lens group and a second lens group are disposed in a second position in an optical imaging system.

(4) FIG. 4 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 3.

(5) FIG. 5 is a diagram illustrating a second example in which a first lens group and a second lens group are disposed in a first position in an optical imaging system.

(6) FIG. 6 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 5.

(7) FIG. 7 is a diagram illustrating a second example in which a first lens group and a second lens group are disposed in a second position in an optical imaging system.

(8) FIG. 8 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 7.

(9) FIG. 9 is a diagram illustrating a third example in which a first lens group and a second lens group are disposed in a first position in an optical imaging system.

(10) FIG. 10 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 9.

- (11) FIG. 11 is a diagram illustrating a third example in which a first lens group and a second lens group are disposed in a second position in an optical imaging system.
- (12) FIG. 12 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 11.
- (13) FIG. 13 is diagram illustrating a fourth example in which a first lens group and a second lens group are disposed in a first position in an optical imaging system.
- (14) FIG. 14 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 13.
- (15) FIG. 15 is a diagram illustrating a fourth example in which a first lens group and a second lens group are disposed in a second position in an optical imaging system.
- (16) FIG. 16 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 15.
- (17) FIG. 17 is a diagram illustrating a fifth example in which a first lens group and a second lens group are disposed in a first position in an optical imaging system.
- (18) FIG. 18 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 17.
- (19) FIG. 19 is a diagram illustrating a fifth example in which a first lens group and a second lens group are disposed in a second position in an optical imaging system.
- (20) FIG. 20 is a curve indicating aberration properties of the optical imaging system illustrated in FIG. 19.
- (21) FIG. 21 is a modified example of a plurality of reflective members according to an example.
- (22) Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depictions of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

- (23) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent to one of ordinary skill in the art. The sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed, as will be apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Also, descriptions of functions and constructions that would be well known to one of ordinary skill in the art may be omitted for increased clarity and conciseness.
- (24) The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to one of ordinary skill in the art.
- (25) Herein, it is to be noted that use of the term “may” with respect to an embodiment or example, e.g., as to what an embodiment or example may include or implement, means that at least one embodiment or example exists in which such a feature is included or implemented while all examples and examples are not limited thereto.
- (26) Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.
- (27) As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items.
- (28) Although terms such as “first,” “second,” and “third” may be used herein to describe various

members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

(29) Spatially relative terms such as “above,” “upper,” “below,” and “lower” may be used herein for ease of description to describe one element's relationship to another element as illustrated in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being “above” or “upper” relative to another element will then be “below” or “lower” relative to the other element. Thus, the term “above” encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

(30) The terminology used herein is for describing various examples only, and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes,” and “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

(31) Due to manufacturing techniques and/or tolerances, variations of the shapes illustrated in the drawings may occur. Thus, the examples described herein are not limited to the specific shapes illustrated in the drawings, but include changes in shape occurring during manufacturing.

(32) The features of the examples described herein may be combined in various manners as will be apparent after gaining an understanding of the disclosure of this application. Further, although the examples described herein have a variety of configurations, other configurations are possible as will be apparent after gaining an understanding of the disclosure of this application.

(33) The drawings may not be to scale, and the relative sizes, proportions, and depictions of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

(34) In the diagrams illustrating the lenses, a thickness, a size, and a shape of the lens are exaggerated to illustrate an example, and a spherical or an aspherical shape of the lens illustrated in the diagram is an example, and a shape is not limited thereto.

(35) The optical imaging system of the various examples may be mounted on a portable electronic device. For example, the optical imaging system may be a component of a camera module mounted in a portable electronic device. The portable electronic device may be implemented as a portable electronic device such as a mobile communication terminal, a smart phone, or a tablet PC.

(36) The optical imaging system may include a plurality of lenses. The plurality of lenses may be spaced apart from each other by a predetermined distance.

(37) As an example, the optical imaging system may include at least six lenses. However, the number of lenses is not limited thereto, and the number of lenses may be smaller or greater than six is desired.

(38) A first lens (or a forwardmost lens) may refer to the lens most adjacent to an object side (or a first reflective member) along an optical axis, and a last lens (or a rearmost lens) refers to a lens most adjacent to an imaging surface (or a second reflective member) along the optical axis.

(39) Also, in each lens, a first surface may refer to a surface adjacent to an object side (or an object-side surface), and a second surface may refer to a surface adjacent to an image side (or an image-side surface). Also, in the example embodiment, a radius of curvature, a thickness, a distance, and a focal length of the lens may be represented in mm, and a unit of a field of view (FOV) may be degree.

- (40) The imaging surface may refer to a virtual surface on which a focus is formed by the optical imaging system. Alternatively, the imaging surface may refer to one surface of an image sensor through which light is received.
- (41) The optical imaging system may include at least six lenses.
- (42) For example, the optical imaging system may include a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens arranged in order from an object side (or the first reflective member).
- (43) The optical imaging system may include seven or more lenses if desired. Also, the optical imaging system may further include an image sensor for converting an incident image of a subject into an electric signal.
- (44) The optical imaging system may further include a plurality of reflective members having reflective surfaces which may change an optical path. For example, each of the plurality of reflective members may be a mirror or a prism.
- (45) One of the plurality of reflective members may be disposed on a front side of the plurality of lenses. For example, the first reflective member may be disposed on a front side of the first lens (disposed adjacent to the object side than the first lens). The other reflective members may be disposed on a rear side of the plurality of lenses. As an example, the other reflective member may be disposed between the sixth lens and the image sensor (or an imaging surface) along the optical axis.
- (46) By refracting an optical path several times by arranging the plurality of reflective members between the plurality of lenses and the image sensor, the light path may be formed long in a relatively narrow space.
- (47) Accordingly, the optical imaging system may have a reduced size and a long focal length.
- (48) Also, the optical imaging system may further include an infrared cut-off filter (hereinafter, referred to as a filter) for blocking infrared rays. The filter may be disposed between the reflective member disposed most adjacent to the image sensor among the plurality of reflective members and the image sensor.
- (49) Overall lenses included in the optical imaging system in an example embodiment may be formed of a plastic material.
- (50) One of the plurality of lenses may be configured as a lens of which a focal length changes (hereinafter, referred to as a “variable lens”). The variable lens may include a liquid lens and a flat lens attached to the liquid lens.
- (51) The lenses other than the variable lens may have an aspherical surface. For example, each of the lenses other than the variable lens may have at least one aspherical surface.
- (52) The aspherical surface of the lens may be represented by Equation 1.
- (53)
$$Z = \frac{cY^2}{1 + \sqrt{1 - (1 + K)c^2Y^2}} + AY^4 + BY^6 + CY^8 + DY^{10} + EY^{12} \quad [\text{Equation 1}]$$
- (54) In Equation 1, c is a radius of curvature of the lens (a reciprocal of a radius of curvature), K is a conic constant, and Y is a distance from an arbitrary point on the aspherical surface of the lens to the optical axis. Also, constants A to E refer to aspheric coefficients. Z represents a distance (SAG) from an arbitrary point on the aspherical surface of the lens to a vertex of the aspherical surface.
- (55) The optical imaging system may include a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens arranged in order from the object side toward the image side. The second lens or the sixth lens may be a variable lens.
- (56) For example, the second lens or the sixth lens may have positive refractive power or negative refractive power as the focal length changes.
- (57) In the optical imaging system, since a focal length of one of the plurality of lenses may change, the focal length of the optical imaging system may also change.
- (58) Therefore, even when the lens is not directly moved, the focal length of the optical imaging system may change, such that a zoom effect may be easily implemented.

(59) When the focal length is changed by directly moving the lens, a moving distance of the lens may need to be secured, and accordingly, the overall length of the optical imaging system may increase. However, the optical imaging system in an example embodiment may implement a zoom effect and may have a reduced size.

(60) The variable lens may include a liquid lens and a flat lens.

(61) In the liquid lens, a radius of curvature and a thickness of the object-side surface may change. Accordingly, the focal length of the liquid lens may change. The thickness may refer to the thickness on the optical axis.

(62) When the variable lens has positive refractive power, the radius of curvature of the object-side surface of the liquid lens may be positive. When the variable lens has negative refractive power, the radius of curvature of the object-side surface of the liquid lens may be negative.

(63) The thickness on the optical axis of the liquid lens may be smaller when the variable lens has negative refractive power than when the variable lens has positive refractive power.

(64) The flat lens may be coupled to an image-side surface of the liquid lens to support the liquid lens. For example, the liquid lens and the flat lens may be bonded to each other. In the flat lens, both the object-side surface and the image-side surface may be flat.

(65) A plurality of reflective members may be disposed between the plurality of lenses and the image sensor. For example, the plurality of reflective members may be disposed between a rearmost lens among the plurality of lenses and the image sensor. The optical axis may be folded by the plurality of reflective members disposed between the rearmost lens and the image sensor.

(66) The plurality of reflective members disposed between the rearmost lens and the image sensor may be moved. Accordingly, a distance on the optical axis between the rearmost lens and the image sensor may change, such that, even when focal length of the imaging optical imaging system changes, a focus may be easily formed on the image sensor (or the imaging surface).

(67) Accordingly, the optical imaging system may have an optical zoom function and a focus adjustment function.

(68) The optical imaging system may have characteristics of a telephoto lens having a relatively narrow field of view and a long focal length.

(69) The optical imaging system may satisfy at least one of conditional equations as below:

$$10 < fv_1/D6R_1 \quad [\text{Conditional equation 1}]$$

$$-15 < fv_2/D6R_2 < -3 \quad [\text{Conditional equation 2}]$$

$$-7 < (fv_1 \times D6R_2)/(fv_2 \times D6R_1) < -1 \quad [\text{Conditional equation 3}]$$

$$-3 < fv_2/fv_1 < 0 \quad [\text{Conditional equation 4}]$$

$$3 < D6R_2/D6R_1 < 6 \quad [\text{Conditional equation 5}]$$

$$0 < L1/TTL1 < 1 \quad [\text{Conditional equation 6}]$$

$$0 < L2/TTL2 < 1 \quad [\text{Conditional equation 7}]$$

$$1 < (L1 \times TTL2)/(L2 \times TTL1) < 3 \quad [\text{Conditional equation 8}]$$

(70) In the conditional equations, fv_1 is a first focal length of the variable lens, and fv_2 is a second focal length of the variable lens.

(71) A focal length of the variable lens may change. Accordingly, the focal length of the optical imaging system may also change. For example, the optical imaging system may have a first focal length when the variable lens has a first focal length, and the optical imaging system may have a second focal length when the variable lens has a second focal length.

(72) The first focal length of the variable lens may be positive, and the second focal length may be negative.

(73) The second focal length of the optical imaging system may be greater than the first focal length.

(74) $D6R_1$ is a distance on the optical axis from the rearmost lens to the second reflective member when the optical imaging system has the first focal length (or the variable lens has the first focal length).

(75) D6R_2 is a distance on the optical axis from the rearmost lens to the second reflective member when the optical imaging system has a second focal length (or when the variable lens has a second focal length).

(76) When the optical imaging system may include six lenses, the rearmost lens may be a sixth lens.

(77) L1 is a linear distance in a direction parallel to the optical axis from the object-side surface of the first lens to the imaging surface when the imaging optical imaging system has the first focal length (or when the variable lens has the first focal length) to be.

(78) L2 is a linear distance in a direction parallel to the optical axis from the object-side surface of the first lens to the imaging surface when the imaging optical imaging system has a second focal length (or when the variable lens has a second focal length). L1 and L2 may be the same.

(79) TTL1 is a distance on the optical axis from the object-side surface of the first lens to the imaging surface when the optical imaging system has the first focal length (or the variable lens has the first focal length).

(80) TTL2 is a distance on the optical axis from the object-side surface of the first lens to the imaging surface when the optical imaging system has the second focal length (or the variable lens has the second focal length).

(81) A first example of the optical imaging system will be described with reference to FIGS. 1 to 4.

(82) FIG. 1 is a diagram illustrating a state in which the optical imaging system has a first focal length, and FIG. 3 is a diagram illustrating a state in which the optical imaging system has a second focal length.

(83) The optical imaging system in a first example may include a first lens **110**, a second lens **120**, a third lens **130**, a fourth lens **140**, a fifth lens **150**, and a sixth lens **160**.

(84) The optical imaging system may form a focus on an imaging surface **180**.

(85) Also, the optical imaging system may further include a filter **170** and an image sensor IS. Also, a stop may be disposed between the third lens **130** and the fourth lens **140**.

(86) Also, the optical imaging system may further include a first reflective member R1 disposed on a front side (object side) of the first lens **110** and having a reflective surface for changing an optical path. The first reflective member R1 may be implemented by a prism, or may be provided as a mirror.

(87) In FIGS. 1 and 3, the prism, the first reflective member R1, may have a flat plate shape, but the actual prism may have a triangular columnar shape.

(88) Light incident to the first reflective member R1 may be refracted by the first reflective member R1 and may pass through the first to sixth lenses **110** to **160**. For example, light incident on the first reflective member R1 along the first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

(89) Also, the optical imaging system may further include a second reflective member R2 and a third reflective member R3 disposed on a rear side (image side) of the sixth lens **160** and each having a reflective surface for changing an optical path. The second reflective member R2 and the third reflective member R3 may be disposed between the sixth lens **160** and the image sensor IS.

(90) In the first example, the second reflective member R2 and the third reflective member R3 may be implemented as mirrors, or may be provided as prisms. Also, as illustrated in FIG. 21, the second reflective member R2 and the third reflective member R3 may be provided in the form of a single prism R2' having two reflective surfaces.

(91) Light passing through the first to sixth lenses **110** to **160** may be refracted by the second reflective member R2. For example, light passing through the first to sixth lenses **110** to **160** may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflective member R2. The second reflective member R2 may have a reflective surface for refracting light to a third optical axis perpendicular to the second optical axis.

(92) Light refracted by the second reflective member R2 may be refracted again by the third

reflective member R3. For example, light refracted toward the third optical axis by the second reflective member R2 may be refracted toward the fourth optical axis perpendicular to the third optical axis by the third reflective member R3, and may be received in the image sensor IS. The third reflective member R3 may have a reflective surface for refracting light to a fourth optical axis perpendicular to the third optical axis.

(93) The second optical axis and the fourth optical axis may be parallel to each other.

(94) The filter **170** may be disposed between the third reflective member R3 and the image sensor IS.

(95) At least one of the first to sixth lenses **110** to **160** may be a variable lens having a focal length which may change. For example, the second lens **120** may be a variable lens. The second lens **120** may include a liquid lens **121** and a flat lens **123** attached to an image-side surface of the liquid lens **121**.

(96) A radius of curvature and a thickness of the object-side surface of the liquid lens **121** may change. Accordingly, the focal length of the liquid lens **121** may change. The thickness may refer to the thickness on the optical axis.

(97) As the focal length of the second lens **120** changes, the focal length of the optical imaging system may also change.

(98) Even when the first to sixth lenses **110** to **160** are not moved, the focal length of the imaging optical imaging system may change by changing the focal length of the second lens **120**, such that the zoom effect may be easily implemented.

(99) Since the focal length of the optical imaging system changes, the second reflective member R2 and the third reflective member R3 may move to form a focus on the imaging surface **180**. For example, as the second reflective member R2 and the third reflective member R3 move, a distance between the sixth lens **160** and the second reflective member R2 may change, and a distance between the third reflective member R3 and the image sensor IS (or the imaging surface **180**) may also change.

(100) Here, the “distance” may refer to the distance on the optical axis between the two members.

(101) The second reflective member R2 and the third reflective member R3 may be moved together. Accordingly, a distance on the optical axis between the second reflective member R2 and the third reflective member R3 may not change.

(102) The length of the optical path may change by the movement of the second reflective member R2 and the third reflective member R3.

(103) The first reflective member R1 and the first to sixth lenses **110** to **160** may be fixed members of which positions may be fixed, and the second reflective member R2 and the third reflective member R3 may be movable members of which positions may change.

(104) Characteristics of each lens (a radius of curvature, a thickness of the lens or a distance between lenses, a refractive index, and an Abbe number) may be listed in Table 1.

(105) In Table 1, the refractive indexes of the second reflective member R2 and the third reflective member R3 may be indicated by negative numbers, which may indicate that the second reflective member R2 and the third reflective member R3 may have a reflective surface for reflecting light.

(106) TABLE-US-00001

	TABLE 1	Radius	Thickness	Surface of	or Refractive	Abbe No.
Note	curvature	distance	index	number	S1	First
	Infinity	2.4500	1.722	29.5	reflective member	S2
	Infinity	2.4500	1.722	29.5	S3	Infinity
	1.9421	S4	First	-4.9337	2.0000	1.671
	20	S5	lens	-5.4289	D12	S6
	Second	R3	T2	1.298	100	S7
	lens	Infinity	0.5000	1.526	55	S8
	Infinity	2.0000	S9	Third	-5.1925	0.9800
	1.544	56	S10	lens	26.6662	0.0940
	S11	Stop	Infinity	0.1000	S12	Fourth
	-2804.6225	1.0000	1.544	56	S13	lens
	-6.5474	0.1000	S14	Fifth	21.9524	1.0000
	1.595	31.1	S15	lens	5.3446	0.1000
	S16	Sixth	4.7880	1.4165	1.544	56
	S17	lens	-25.9774	D6R	S18	Second
	Infinity	0.0000	-1	reflective member	S19	Infinity
	-7.0000	S20	Third	Infinity	0.0000	-1
	reflective member	S21	Infinity	DR3F	S22	Filter
	Infinity	0.2100	1.519	64.2	S23	Infinity
	2.0350	S24	Imaging	Infinity	surface	

(107) TABLE-US-00002 TABLE 2 In case of first In case of second focal length focal length R3
10.8963 -22.0525 D12 0.7278 1.1606 T2 2.2907 1.8578 D6R 3 17.0064 DR3F 5.0347 19.0410

(108) R3 is a radius of curvature of the object-side surface of the second lens **120**, D12 is a distance on the optical axis from the image-side surface of the first lens **110** to the object-side surface of the second lens **120**, T2 is the thickness on the optical axis of the second lens **120**, D6R is the distance on the optical axis from the image-side surface of the sixth lens **160** to the second reflective member R2, and DR3F is the distance on the optical axis between the third reflective member R3 and the filter **170**.

(109) TABLE-US-00003 TABLE 3 In case of first In case of second focal length focal length f
16.2019 31.7632 f1 129.4498 129.4498 f2 36.5646 -74.0016 f3 -7.9005 -7.9005 f4 12.0602
12.0602 f5 -12.1492 -12.1492 f6 7.5529 7.5529

(110) f is the focal length of the optical imaging system, f1 is the focal length of the first lens, f2 is the focal length of the second lens, f3 is the focal length of the third lens, f4 is the focal length of the fourth lens, f5 is the focal length of the fifth lens, and f6 is the focal length of the sixth lens.

(111) In the first example, the first lens **110** may have positive refractive power, the first surface of the first lens **110** may be concave, and the second surface of the first lens **110** may be convex.

(112) The second lens **120** may be a variable lens of which the focal length may change. For example, the second lens **120** may have a first focal length and a second focal length. The second lens **120** may include a liquid lens **121** and a flat lens **123** attached to an image-side surface of the liquid lens **121**.

(113) When the second lens **120** has a first focal length, the second lens **120** may have positive refractive power. Also, the first surface of the second lens **120** (e.g., the object-side surface of the liquid lens **121**) may have a convex, and the second surface of the second lens **120** (e.g., the image side of the flat lens **123**) may be flat.

(114) When the second lens **120** has a second focal length, the second lens **120** may have negative refractive power. Also, the first surface of the second lens **120** may be concave, and the second surface of the second lens **120** may be flat.

(115) The third lens **130** may have negative refractive power, and the first and second surfaces of the third lens **130** may be concave.

(116) The fourth lens **140** may have positive refractive power, the first surface of the fourth lens **140** may be concave, and the second surface of the fourth lens **140** may be convex.

(117) The fifth lens **150** may have negative refractive power, the first surface of the fifth lens **150** may be convex, and the second surface of the fifth lens **150** may be concave.

(118) The sixth lens **160** may have positive refractive power, and the first and second surfaces of the sixth lens **160** may be convex.

(119) Each surface of the first lens **110** and the third lens **130** to the sixth lens **160** may have an aspherical coefficient as illustrated in Table 8. For example, the object-side surfaces and the image-side surfaces of the first lens **110** and the third lens **130** to sixth lens **160** may be aspherical.

(120) TABLE-US-00004 TABLE 4 Conic Fourth Sixth Eighth Tenth Twelfth constant coefficient
coefficient coefficient coefficient coefficient (K) (A) (B) (C) (D) (E) S4 0 0.000485033 5.92E-05
-1.34E-06 -8.69E-09 1.53E-08 S5 0 0.000531678 3.41E-05 -6.41E-07 -1.56E-08 6.36E-09
S9 0 0.002786538 -1.61E-05 2.50E-06 -1.30E-09 1.41E-07 S10 0 -0.000184995 -1.88E-06
3.42E-06 1.64E-07 1.25E-07 S12 0 -0.000463124 -2.94E-05 -5.03E-06 -6.15E-07
-4.74E-08 S13 0 0.000766595 -3.87E-06 -5.50E-07 -5.64E-07 -9.26E-08 S14 0
-2.58E-04 2.75E-05 2.71E-06 -8.01E-08 6.64E-09 S15 0 2.89641E-05 -4.24E-05
-8.80E-06 -4.67E-07 -6.17E-08 S16 0 -0.000543734 -4.28E-05 -3.16E-06 1.06E-07
2.91E-08 S17 0 -0.00027459 3.69E-05 3.07E-06 8.39E-07 7.03E-08

(121) Also, the optical imaging system configured as described above may have aberration properties illustrated in FIGS. 2 and 4.

(122) A second example of the optical imaging system will be described with reference to FIGS. 5

to 8.

(123) FIG. 5 is a diagram illustrating a state in which the optical imaging system has a first focal length, and FIG. 7 is a diagram illustrating a state in which the optical imaging system has a second focal length.

(124) The optical imaging system in a second example may include a first lens **210**, a second lens **220**, a third lens **230**, a fourth lens **240**, a fifth lens **250**, and a sixth lens **260**.

(125) The optical imaging system may form a focus on an imaging surface **280**.

(126) Also, the optical imaging system may further include a filter **270** and an image sensor IS. Also, a stop may be disposed between the third lens **230** and the fourth lens **240**.

(127) Also, the optical imaging system may further include a first reflective member R1 disposed on a front side of the first lens **210** and having a reflective surface for changing an optical path. The first reflective member R1 may be implemented by a prism, or may be provided as a mirror.

(128) In FIGS. 5 and 7, the prism, the first reflective member R1, may have a flat plate shape, but the actual prism may have a triangular columnar shape.

(129) Light incident to the first reflective member R1 may be refracted by the first reflective member R1 and may pass through the first to sixth lenses **210** to **260**. For example, light incident to the first reflective member R1 along the first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

(130) Also, the optical imaging system may further include a second reflective member R2 and a third reflective member R3 disposed on a rear side of the sixth lens **260** and each having a reflective surface for changing an optical path. The second reflective member R2 and the third reflective member R3 may be disposed between the sixth lens **260** and the image sensor IS.

(131) The second reflective member R2 and the third reflective member R3 may be implemented as mirrors, or may be provided as prisms. Also, as illustrated in FIG. 21, the second reflective member R2 and the third reflective member R3 may be provided in the form of a single prism R2' having two reflective surfaces.

(132) Light passing through the first to sixth lenses **210** to **260** may be refracted by the second reflective member R2. For example, light passing through the first to sixth lenses **210** to **260** may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflective member R2. The second reflective member R2 may have a reflective surface for refracting light to a third optical axis perpendicular to the second optical axis.

(133) Light refracted by the second reflective member R2 may be refracted again by the third reflective member R3. For example, light refracted toward the third optical axis by the second reflective member R2 may be refracted toward the fourth optical axis perpendicular to the third optical axis by the third reflective member R3, and may be received in the image sensor IS. The third reflective member R3 may have a reflective surface for refracting light to a fourth optical axis perpendicular to the third optical axis.

(134) The second optical axis and the fourth optical axis may be parallel to each other.

(135) The filter **270** may be disposed between the third reflective member R3 and the image sensor IS.

(136) At least one of the first to sixth lenses **210** to **260** may be a variable lens having a focal length which may change. For example, the second lens **220** may be a variable lens. The second lens **220** may include a liquid lens **221** and a flat lens **223** attached to an image-side surface of the liquid lens **221**.

(137) A radius of curvature and a thickness of the object-side surface of the liquid lens **221** may change. Accordingly, the focal length of the liquid lens **221** may change. The thickness may refer to the thickness on the optical axis.

(138) As the focal length of the second lens **220** changes, the focal length of the optical imaging system may also change.

(139) Even when the first to sixth lenses **210** to **260** are not moved, the focal length of the imaging

optical imaging system may change by changing the focal length of the second lens **220**, such that the zoom effect may be easily implemented.

(140) Since the focal length of the optical imaging system changes, the second reflective member R2 and the third reflective member R3 may move to form a focus on the imaging surface. For example, as the second reflective member R2 and the third reflective member R3 move, a distance between the sixth lens **260** and the second reflective member R2 may change, and a distance between the third reflective member R3 and the image sensor IS (or the imaging surface **280**) may also change.

(141) Here, the “distance” may refer to the distance on the optical axis between the two members.

(142) The second reflective member R2 and the third reflective member R3 may be moved together. Accordingly, a distance on the optical axis between the second reflective member R2 and the third reflective member R3 may not change.

(143) The length of the optical path may change by the movement of the second reflective member R2 and the third reflective member R3.

(144) The first reflective member R1 and the first to sixth lenses **210** to **260** may be fixed members of which positions may be fixed, and the second reflective member R2 and the third reflective member R3 may be movable members of which positions may change.

(145) Characteristics of each lens (a radius of curvature, a thickness of the lens or a distance between lenses, a refractive index, and an Abbe number) may be listed in Table 5.

(146) In Table 5, the refractive indexes of the second reflective member R2 and the third reflective member R3 may be indicated by negative numbers, which may indicate that the second reflective member R2 and the third reflective member R3 may have a reflective surface for reflecting light.

(147) TABLE-US-00005 TABLE 5 Radius Thickness Surface of or Refractive Abbe No. Note curvature distance index number S1 First Infinity 2.4400 1.722 29.5 reflective member S2 Infinity 2.4400 1.722 29.5 S3 Infinity 1.9421 S4 First -10.1200 2.0000 1.671 20 S5 lens -10.3100 D12 S6 Second R3 T2 1.298 100 S7 lens Infinity 0.5000 1.526 55 S8 Infinity 2.0000 S9 Third -6.7700 1.0000 1.544 56 S10 lens 10.5300 0.0950 S11 Stop Infinity 0.1000 S12 Fourth 14.5000 1.0000 1.544 56 S13 lens -7.2000 0.1000 S14 Fifth 366.6400 1.0000 1.595 31.1 S15 lens 7.3600 0.1000 S16 Sixth 6.0400 1.4369 1.544 56 S17 lens -17.1965 D6R S18 Second Infinity 0.0000 -1 reflective member S19 Infinity -7.3358 S20 Third Infinity 0.0000 -1 reflective member S21 Infinity DR3F S22 Filter Infinity 0.2100 1.519 64.2 S23 Infinity 1.6100 S24 Imaging Infinity surface

(148) TABLE-US-00006 TABLE 6 In case of first In case of second focal length focal length R3 14.19 -16.246 D12 1.3194 1.7348 T2 2.222 1.8065 D6R 3 13.6968 DR3F 5 15.6968

(149) The definitions of R3, D12, T2, D6R and DR3F may be the same as in the first example.

(150) TABLE-US-00007 TABLE 7 In case of first In case of second focal length focal length f 16.58 28.55 f1 250 250 f2 47.6 -54.5168 f3 -7.42 -7.42 f4 9 9 f5 -12.651 -12.651 f6 8.4 8.4

(151) The definitions of f, f1, f2, f3, f4, f5, and f6 may be the same as those of the first example.

(152) In the second example, the first lens **210** may have positive refractive power, the first surface of the first lens **210** may be concave, and the second surface of the first lens **210** may be convex.

(153) The second lens **220** may be a variable lens of which the focal length may change. For example, the second lens **220** may have a first focal length and a second focal length. The second lens **220** may include a liquid lens **221** and a flat lens **223** attached to an image-side surface of the liquid lens **221**.

(154) When the second lens **220** has a first focal length, the second lens **220** may have positive refractive power. Also, the first surface of the second lens **220** (e.g., the object-side surface of the liquid lens **221**) may be convex, and the second surface of the second lens **220** (e.g., the image side of the flat lens **223**) may be flat.

(155) When the second lens **220** has a second focal length, the second lens **220** may have negative refractive power. Also, the first surface of the second lens **220** may be concave, and the second

surface of the second lens **220** may be flat.

(156) The third lens **230** may have negative refractive power, and the first and second surfaces of the third lens **230** may be concave.

(157) The fourth lens **240** may have positive refractive power, the first surface of the fourth lens **240** may be concave, and the second surface of the fourth lens **240** may be convex.

(158) The fifth lens **250** may have negative refractive power, the first surface of the fifth lens **250** may be convex, and the second surface of the fifth lens **250** may be concave.

(159) The sixth lens **260** may have positive refractive power, and the first and second surfaces of the sixth lens **260** may be convex.

(160) Each surface of the first lens **210** and the third lens **230** to the sixth lens **260** may have an aspherical coefficient as illustrated in Table 8. For example, the object-side surfaces and the image-side surfaces of the first lens **210** and the third lens **230** to sixth lens **260** may be aspherical.

(161) TABLE-US-00008 TABLE 8 Conic Fourth Sixth Eighth Tenth Twelfth constant coefficient coefficient coefficient coefficient (K) (A) (B) (C) (D) (E) S4 0 0.001045263 6.30E-05 -2.11E-05 1.92E-06 -7.83E-08 S5 0 0.001497638 -1.86E-05 -1.21E-06 -5.65E-07 3.85E-08 S9 0 0.005401982 -4.42E-04 1.54E-08 6.99E-06 -1.24E-06 S10 0 -0.000320078 1.01E-05 -9.72E-06 -4.50E-06 -6.99E-07 S12 0 -0.000949189 -5.73E-05 -1.06E-05 -1.84E-06 -7.99E-08 S13 0 0.002971337 5.63E-07 2.15E-06 1.99E-06 -2.66E-07 S14 0 8.08E-04 2.22E-04 9.59E-06 -2.39E-06 -5.87E-07 S15 0 -0.000245536 -1.15E-04 -1.22E-05 -8.45E-07 1.66E-08 S16 0 -0.002447921 -1.19E-04 -6.54E-06 1.03E-06 6.85E-08 S17 0 -0.001864786 4.16E-05 1.19E-05 -1.68E-06 1.50E-07

(162) Also, the optical imaging system configured as described above may have aberration properties illustrated in FIGS. **6** and **8**.

(163) A third example of the optical imaging system will be described with reference to FIGS. **9** to **12**.

(164) FIG. **9** is a diagram illustrating a state in which the optical imaging system has a first focal length, and FIG. **11** is a diagram illustrating a state in which the optical imaging system has a second focal length.

(165) The optical imaging system in a third example may include a first lens **310**, a second lens **320**, a third lens **330**, a fourth lens **340**, a fifth lens **350**, and a sixth lens **360**.

(166) The optical imaging system in the third example may form a focus on an imaging surface **380**.

(167) Also, the optical imaging system may further include a filter **370** and an image sensor IS. Also, a stop may be disposed between the third lens **330** and the fourth lens **340**.

(168) Also, the optical imaging system may further include a first reflective member R1 disposed on a front side of the first lens **310** and having a reflective surface for changing an optical path. The first reflective member R1 may be implemented by a prism, or may be provided as a mirror.

(169) In FIGS. **9** and **11**, the prism, the first reflective member R1, may have a flat plate shape, but the actual prism may have a triangular columnar shape.

(170) Light incident to the first reflective member R1 may be refracted by the first reflective member R1 and may pass through the first to sixth lenses **310** to **360**. For example, light incident to the first reflective member R1 along the first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

(171) Also, the optical imaging system may further include a second reflective member R2 and a third reflective member R3 disposed on a rear side of the sixth lens **360** and each having a reflective surface for changing an optical path. The second reflective member R2 and the third reflective member R3 may be disposed between the sixth lens **360** and the image sensor IS.

(172) The second reflective member R2 and the third reflective member R3 may be implemented as mirrors, or may be provided as prisms. Also, as illustrated in FIG. **21**, the second reflective member R2 and the third reflective member R3 may be provided in the form of a single prism R2'

having two reflective surfaces.

(173) Light passing through the first to sixth lenses **310** to **360** may be refracted by the second reflective member R2. For example, light passing through the first to sixth lenses **310** to **360** may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflective member R2. The second reflective member R2 may have a reflective surface for refracting light to a third optical axis perpendicular to the second optical axis.

(174) Light refracted by the second reflective member R2 may be refracted again by the third reflective member R3. For example, light refracted toward the third optical axis by the second reflective member R2 may be refracted toward the fourth optical axis perpendicular to the third optical axis by the third reflective member R3, and may be received in the image sensor IS. The third reflective member R3 may have a reflective surface for refracting light to a fourth optical axis perpendicular to the third optical axis.

(175) The second optical axis and the fourth optical axis may be parallel to each other.

(176) The filter **370** may be disposed between the third reflective member R3 and the image sensor IS.

(177) At least one of the first to sixth lenses **310** to **360** may be a variable lens having a focal length which may change. For example, the second lens **320** may be a variable lens. The second lens **320** may include a liquid lens **321** and a flat lens **323** attached to an image-side surface of the liquid lens **321**.

(178) A radius of curvature and a thickness of the object-side surface of the liquid lens **321** may change. Accordingly, the focal length of the liquid lens **321** may change. The thickness may refer to the thickness on the optical axis.

(179) As the focal length of the second lens **320** changes, the focal length of the optical imaging system may also change.

(180) Even when the first to sixth lenses **310** to **360** are not moved, the focal length of the imaging optical imaging system may change by changing the focal length of the second lens **320**, such that the zoom effect may be easily implemented.

(181) Since the focal length of the optical imaging system changes, the second reflective member R2 and the third reflective member R3 may move to form a focus on the imaging surface. For example, as the second reflective member R2 and the third reflective member R3 move, a distance between the sixth lens **360** and the second reflective member R2 may change, and a distance between the third reflective member R3 and the image sensor IS (or the imaging surface **380**) may also change.

(182) Here, the “distance” may refer to the distance on the optical axis between the two members.

(183) The second reflective member R2 and the third reflective member R3 may be moved together. Accordingly, a distance on the optical axis between the second reflective member R2 and the third reflective member R3 may not change.

(184) The length of the optical path may change by the movement of the second reflective member R2 and the third reflective member R3.

(185) The first reflective member R1 and the first to sixth lenses **310** to **360** may be fixed members of which positions may be fixed, and the second reflective member R2 and the third reflective member R3 may be movable members of which positions may change.

(186) Characteristics of each lens (a radius of curvature, a thickness of the lens or a distance between lenses, a refractive index, and an Abbe number) may be listed in Table 9.

(187) In Table 9, the refractive indexes of the second reflective member R2 and the third reflective member R3 may be indicated by negative numbers, which may indicate that the second reflective member R2 and the third reflective member R3 may have a reflective surface for reflecting light.

(188) TABLE-US-00009 TABLE 9 Radius Thickness Surface of or Refractive Abbe No. Note curvature distance index number S1 First Infinity 2.4300 1.722 29.5 reflective member S2 Infinity 2.4300 1.722 29.5 S3 Infinity 1.9421 S4 First -5.4127 2.0000 1.671 20 S5 lens -5.8461 D12 S6

Second R3 T2 1.298 100 S7 lens Infinity 0.5000 1.526 55 S8 Infinity 2.0000 S9 Third -6.7841
 1.0000 1.544 56 S10 lens 10.4390 0.0941 S11 Stop Infinity 0.1000 S12 Fourth 14.2659 1.0000
 1.544 56 S13 lens -7.0225 0.1000 S14 Fifth 1176.7971 1.0000 1.595 31.1 S15 lens 7.1031 0.1000
 S16 Sixth 5.8531 1.4165 1.544 56 S17 lens -20.1261 D6R S18 Second Infinity 0.0000 -1
 reflective member S19 Infinity -7.0000 S20 Third Infinity 0.0000 -1 reflective member S21
 Infinity DR3F S22 Filter Infinity 0.2100 1.519 64.2 S23 Infinity 2.1834 S24 Imaging Infinity
 surface

(189) TABLE-US-00010 TABLE 10 In case of first In case of second focal length focal length R3
 17.709 -16.505 D12 1.0147 1.3824 T2 2.1774 1.8096 D6R 3 13.9994 DR3F 6.1990 17.1984

(190) The definitions of R3, D12, T2, D6R and DR3F may be the same as in the first example.

(191) TABLE-US-00011 TABLE 11 In case of first In case of second focal length focal length f
 16.8083 28.9971 f1 127.387693 127.387693 f2 59.4261745 -55.38590604 f3 -7.40568 -7.40568
 f4 8.7946 8.7946 f5 -12.01742 -12.01742 f6 8.50535 8.50535

(192) The definitions of f, f1, f2, f3, f4, f5, and f6 may be the same as those of the first example.

(193) In the third example, the first lens **310** may have positive refractive power, the first surface of
 the first lens **310** may be concave, and the second surface of the first lens **310** may be convex.

(194) The second lens **320** may be a variable lens of which the focal length may change. For
 example, the second lens **320** may have a first focal length and a second focal length. The second
 lens **320** may include a liquid lens **321** and a flat lens **323** attached to an image-side surface of the
 liquid lens **321**.

(195) When the second lens **320** has a first focal length, the second lens **320** may have positive
 refractive power. Also, the first surface of the second lens **320** (e.g., the object-side surface of the
 liquid lens **321**) may be convex, and the second surface of the second lens **320** (e.g., the image side
 of the flat lens **323**) may be flat.

(196) When the second lens **320** has a second focal length, the second lens **320** may have negative
 refractive power. Also, the first surface of the second lens **320** may be concave, and the second
 surface of the second lens **320** may be flat.

(197) The third lens **330** may have negative refractive power, and the first and second surfaces of
 the third lens **330** may be concave.

(198) The fourth lens **340** may have positive refractive power, the first and second surfaces of the
 fourth lens **340** may be convex.

(199) The fifth lens **350** may have negative refractive power, the first surface of the fifth lens **350**
 may be convex, and the second surface of the fifth lens **350** may be concave.

(200) The sixth lens **360** may have positive refractive power, and the first and second surfaces of
 the sixth lens **360** may be convex.

(201) Each surface of the first lens **310** and the third lens **330** to the sixth lens **360** may have an
 aspherical coefficient as illustrated in Table 12. For example, the object-side surfaces and the
 image-side surfaces of the first lens **310** and the third lens **330** to sixth lens **360** may be aspherical.

(202) TABLE-US-00012 TABLE 12 Conic Fourth Sixth Eighth Tenth Twelfth constant coefficient
 coefficient coefficient coefficient coefficient (K) (A) (B) (C) (D) (E) S4 0 0.000966697 3.73E-05
 1.18E-05 -1.31E-06 3.00E-08 S5 0 0.001076464 3.37E-05 3.97E-06 -2.54E-07 -6.51E-09
 S9 0 0.003884349 2.92E-05 4.69E-06 -1.96E-06 2.03E-08 S10 0 0.000156127 3.51E-05
 2.73E-06 -2.26E-07 -4.52E-08 S12 0 -0.000928506 -7.13E-05 -4.84E-06 -4.12E-07
 -2.81E-08 S13 0 0.001521104 3.09E-05 7.87E-07 -7.52E-07 -1.77E-07 S14 0 -2.14E-07
 7.58E-05 5.53E-06 -1.58E-07 -9.24E-08 S15 0 -0.000186781 -6.07E-05 -6.72E-06
 -5.12E-08 6.80E-08 S16 0 -0.000727204 -4.63E-05 -3.13E-06 3.59E-07 1.51E-09 S17
 0 -0.000664569 2.93E-05 -7.26E-07 1.17E-06 -9.47E-08

(203) Also, the optical imaging system configured as described above may have aberration
 properties illustrated in FIGS. **10** and **12**.

(204) A fourth example of the optical imaging system will be described with reference to FIGS. **13**

to **16**.

(205) FIG. **13** is a diagram illustrating a state in which the optical imaging system has a first focal length, and FIG. **15** is a diagram illustrating a state in which the optical imaging system has a second focal length.

(206) The optical imaging system in a fourth example may include a first lens **410**, a second lens **420**, a third lens **430**, a fourth lens **440**, a fifth lens **450**, and a sixth lens **460**.

(207) The optical imaging system in the fourth example may form a focus on an imaging surface **480**.

(208) Also, the optical imaging system may further include a filter **470** and an image sensor IS. Also, a stop may be disposed between the third lens **430** and the fourth lens **340**.

(209) Also, the optical imaging system may further include a first reflective member R1 disposed on a front side of the first lens **410** and having a reflective surface for changing an optical path. In the fourth example, the first reflective member R1 may be implemented by a prism, or may be provided as a mirror.

(210) In FIGS. **13** and **15**, the prism, the first reflective member R1, may have a flat plate shape, but the actual prism may have a triangular columnar shape.

(211) Light incident to the first reflective member R1 may be refracted by the first reflective member R1 and may pass through the first to sixth lenses **410** to **460**. For example, light incident to the first reflective member R1 along the first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

(212) Also, the optical imaging system may further include a second reflective member R2 and a third reflective member R3 disposed on a rear side of the sixth lens **460** and each having a reflective surface for changing an optical path. The second reflective member R2 and the third reflective member R3 may be disposed between the sixth lens **460** and the image sensor IS.

(213) The second reflective member R2 and the third reflective member R3 may be implemented as mirrors, or may be provided as prisms. Also, as illustrated in FIG. **21**, the second reflective member R2 and the third reflective member R3 may be provided in the form of a single prism R2' having two reflective surfaces.

(214) Light passing through the first to sixth lenses **410** to **460** may be refracted by the second reflective member R2. For example, light passing through the first to sixth lenses **410** to **460** may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflective member R2. The second reflective member R2 may have a reflective surface for refracting light to a third optical axis perpendicular to the second optical axis.

(215) Light refracted by the second reflective member R2 may be refracted again by the third reflective member R3. For example, light refracted toward the third optical axis by the second reflective member R2 may be refracted toward the fourth optical axis perpendicular to the third optical axis by the third reflective member R3, and may be received in the image sensor IS. The third reflective member R3 may have a reflective surface for refracting light to a fourth optical axis perpendicular to the third optical axis.

(216) The second optical axis and the fourth optical axis may be parallel to each other.

(217) The filter **470** may be disposed between the third reflective member R3 and the image sensor IS.

(218) At least one of the first to sixth lenses **410** to **460** may be a variable lens having a focal length which may change. For example, the second lens **420** may be a variable lens. The second lens **420** may include a liquid lens **421** and a flat lens **423** attached to an image-side surface of the liquid lens **421**.

(219) A radius of curvature and a thickness of the object-side surface of the liquid lens **421** may change. Accordingly, the focal length of the liquid lens **421** may change. The thickness may refer to the thickness on the optical axis.

(220) As the focal length of the second lens **420** changes, the focal length of the optical imaging

system may also change.

(221) Even when the first to sixth lenses **410** to **460** are not moved, the focal length of the imaging optical imaging system may change by changing the focal length of the second lens **420**, such that the zoom effect may be easily implemented.

(222) Since the focal length of the optical imaging system changes, the second reflective member R2 and the third reflective member R3 may move to form a focus on the imaging surface. For example, as the second reflective member R2 and the third reflective member R3 move, a distance between the sixth lens **460** and the second reflective member R2 may change, and a distance between the third reflective member R3 and the image sensor IS (or the imaging surface **480**) may also change.

(223) Here, the “distance” may refer to the distance on the optical axis between the two members.

(224) The second reflective member R2 and the third reflective member R3 may be moved together. Accordingly, a distance on the optical axis between the second reflective member R2 and the third reflective member R3 may not change.

(225) The length of the optical path may change by the movement of the second reflective member R2 and the third reflective member R3.

(226) The first reflective member R1 and the first to sixth lenses **410** to **460** may be fixed members of which positions may be fixed, and the second reflective member R2 and the third reflective member R3 may be movable members of which positions may change.

(227) Characteristics of each lens (a radius of curvature, a thickness of the lens or a distance between lenses, a refractive index, and an Abbe number) may be listed in Table 13.

(228) In Table 13, the refractive indexes of the second reflective member R2 and the third reflective member R3 may be indicated by negative numbers, which may indicate that the second reflective member R2 and the third reflective member R3 may have a reflective surface for reflecting light.

(229) TABLE-US-00013 TABLE 13 Radius Thickness Surface of or Refractive Abbe No. Note curvature distance index number S1 First Infinity 2.4500 1.7 29 reflective member S2 Infinity 2.4500 1.7 29 S3 Infinity 1.9421 S4 First -5.8720 2.0000 1.671 20 S5 lens -6.1970 D12 S6 Second R3 T2 1.298 100 S7 lens Infinity 0.5000 1.526 55 S8 Infinity 2.0000 S9 Third -6.8263 1.0000 1.544 56 S10 lens 10.3880 0.0941 S11 Stop Infinity 0.1000 S12 Fourth 14.3128 1.0000 1.544 56 S13 lens -7.0666 0.1000 S14 Fifth lens 308.4550 1.0000 1.595 31.1 S15 7.1680 0.1000 S16 Sixth 5.9102 1.4494 1.544 56 S17 lens -19.9542 D6R S18 Second Infinity 0.0000 -1 reflective member S19 Infinity -7.0000 S20 Third Infinity 0.0000 -1 reflective member S21 Infinity DR3F S22 Filter Infinity 0.2100 1.519 64.2 S23 Infinity 0.901 S24 Imaging Infinity surface

(230) TABLE-US-00014 TABLE 14 In case of first In case of second focal length focal length R3 19.55 -14.52 D12 1.0514 1.4288 T2 2.1605 1.7832 D6R 3 14.054 DR3F 7.3284 18.3824

(231) The definitions of R3, D12, T2, D6R and DR3F may be the same as in the first example.

(232) TABLE-US-00015 TABLE 15 In case of first In case of second focal length focal length f 16.8032 29.1086 f1 113.20688 113.20688 f2 65.60402685 -48.72483221 f3 -7.419199 -7.419199 f4 8.840524 8.840524 f5 -12.352182 -12.352182 f6 8.549017 8.549017

(233) The definitions of f, f1, f2, f3, f4, f5, and f6 may be the same as those of the first example.

(234) In the fourth example, the first lens **410** may have positive refractive power, the first surface of the first lens **410** may be concave, and the second surface of the first lens **410** may be convex.

(235) The second lens **420** may be a variable lens of which the focal length may change. For example, the second lens **420** may have a first focal length and a second focal length. The second lens **420** may include a liquid lens **421** and a flat lens **423** attached to an image-side surface of the liquid lens **421**.

(236) When the second lens **420** has a first focal length, the second lens **420** may have positive refractive power. Also, the first surface of the second lens **420** (e.g., the object-side surface of the liquid lens **421**) may be convex, and the second surface of the second lens **420** (e.g., the image side

of the flat lens **423**) may be flat.

(237) When the second lens **420** has a second focal length, the second lens **420** may have negative refractive power. Also, the first surface of the second lens **420** may be concave, and the second surface of the second lens **420** may be flat.

(238) The third lens **430** may have negative refractive power, and the first and second surfaces of the third lens **430** may be concave.

(239) The fourth lens **440** may have positive refractive power, the first and second surfaces of the fourth lens **440** may be convex.

(240) The fifth lens **450** may have negative refractive power, the first surface of the fifth lens **450** may be convex, and the second surface of the fifth lens **450** may be concave.

(241) The sixth lens **460** may have positive refractive power, and the first and second surfaces of the sixth lens **460** may be convex.

(242) Each surface of the first lens **410** and the third lens **430** to the sixth lens **460** may have an aspherical coefficient as illustrated in Table 16. For example, the object-side surfaces and the image-side surfaces of the first lens **410** and the third lens **430** to sixth lens **460** may be aspherical.

(243) TABLE-US-00016 TABLE 16 Conic Fourth Sixth Eighth Tenth Twelfth constant coefficient coefficient coefficient coefficient (K) (A) (B) (C) (D) (E) S4 0 0.000652593 5.35E-05 -3.79E-06 6.02E-07 -3.07E-08 S5 0 0.000813573 2.75E-05 -6.85E-07 1.72E-07 -8.27E-09 S9 0 0.001658673 -4.07E-05 5.82E-06 -6.04E-07 2.16E-08 S10 0 -0.000244908 -1.22E-05 2.25E-07 -3.68E-07 8.88E-08 S12 0 -0.000306098 -2.19E-05 -3.21E-06 -2.10E-07 -1.01E-08 S13 0 0.000438819 -1.06E-05 1.78E-06 -3.17E-07 -7.66E-08 S14 0 -1.13E-04 2.32E-05 5.83E-07 -1.35E-07 2.95E-08 S15 0 -3.06757E-05 -3.47E-05 -5.57E-06 -6.93E-08 1.84E-08 S16 0 -0.000336051 -2.57E-05 -2.82E-06 7.91E-08 -3.23E-08 S17 0 5.02056E-06 5.45E-05 -7.66E-06 1.59E-06 -1.21E-07

(244) Also, the optical imaging system configured as described above may have aberration properties as illustrated in FIGS. **14** and **16**.

(245) A fifth example of the optical imaging system will be described with reference to FIGS. **17** to **20**.

(246) FIG. **17** is a diagram illustrating a state in which the optical imaging system has a first focal length, and FIG. **19** is a diagram illustrating a state in which the optical imaging system has a second focal length.

(247) The optical imaging system in a fifth example may include a first lens **510**, a second lens **520**, a third lens **530**, a fourth lens **540**, a fifth lens **550**, and a sixth lens **560**.

(248) The optical imaging system in the fifth example may form a focus on an imaging surface **580**.

(249) Also, the optical imaging system may further include a filter **570** and an image sensor IS. Also, a stop may be disposed between the third lens **530** and the fourth lens **540**.

(250) Also, the optical imaging system may further include a first reflective member R1 disposed on a front side of the first lens **510** and having a reflective surface for changing an optical path. In the fifth example, the first reflective member R1 may be implemented by a prism, or may be provided as a mirror.

(251) In FIGS. **17** and **19**, the prism, the first reflective member R1, may have a flat plate shape, but the actual prism may have a triangular columnar shape.

(252) Light incident to the first reflective member R1 may be refracted by the first reflective member R1 and may pass through the first to sixth lenses **510** to **560**. For example, light incident to the first reflective member R1 along the first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

(253) Also, the optical imaging system may further include a second reflective member R2 and a third reflective member R3 disposed on a rear side of the sixth lens **560** and each having a reflective surface for changing an optical path. The second reflective member R2 and the third

reflective member R3 may be disposed between the sixth lens **560** and the image sensor IS.

(254) The second reflective member R2 and the third reflective member R3 may be implemented as mirrors, or may be provided as prisms. Also, as illustrated in FIG. **21**, the second reflective member R2 and the third reflective member R3 may be provided in the form of a single prism R2' having two reflective surfaces.

(255) Light passing through the first to sixth lenses **510** to **560** may be refracted by the second reflective member R2. For example, light passing through the first to sixth lenses **510** to **560** may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflective member R2. The second reflective member R2 may have a reflective surface for refracting light to a third optical axis perpendicular to the second optical axis.

(256) Light refracted by the second reflective member R2 may be refracted again by the third reflective member R3. For example, light refracted toward the third optical axis by the second reflective member R2 may be refracted toward the fourth optical axis perpendicular to the third optical axis by the third reflective member R3, and may be received in the image sensor IS. The third reflective member R3 may have a reflective surface for refracting light to a fourth optical axis perpendicular to the third optical axis.

(257) The second optical axis and the fourth optical axis may be parallel to each other.

(258) The filter **570** may be disposed between the third reflective member R3 and the image sensor IS.

(259) At least one of the first to sixth lenses **510** to **560** may be a variable lens having a focal length which may change. For example, the sixth lens **560** may be a variable lens. The sixth lens **560** may include a liquid lens **561** and a flat lens **563** attached to an image-side surface of the liquid lens **561**.

(260) A radius of curvature and a thickness of the object-side surface of the liquid lens **561** may change. Accordingly, the focal length of the liquid lens **561** may change. The thickness may refer to the thickness on the optical axis.

(261) As the focal length of the sixth lens **560** changes, the focal length of the optical imaging system may also change.

(262) Even when the first to sixth lenses **510** to **560** are not moved, the focal length of the imaging optical imaging system may change by changing the focal length of the sixth lens **560**, such that the zoom effect may be easily implemented.

(263) Since the focal length of the optical imaging system changes, the second reflective member R2 and the third reflective member R3 may move to form a focus on the imaging surface. For example, as the second reflective member R2 and the third reflective member R3 move, a distance between the sixth lens **560** and the second reflective member R2 may change, and a distance between the third reflective member R3 and the image sensor IS (or the imaging surface **580**) may also change.

(264) Here, the "distance" may refer to the distance on the optical axis between the two members.

(265) The second reflective member R2 and the third reflective member R3 may be moved together. Accordingly, a distance on the optical axis between the second reflective member R2 and the third reflective member R3 may not change.

(266) The length of the optical path may change by the movement of the second reflective member R2 and the third reflective member R3.

(267) The first reflective member R1 and the first to sixth lenses **510** to **560** may be fixed members of which positions may be fixed, and the second reflective member R2 and the third reflective member R3 may be movable members of which positions may change.

(268) Characteristics of each lens (a radius of curvature, a thickness of the lens or a distance between lenses, a refractive index, and an Abbe number) may be listed in Table 17.

(269) In Table 17, the refractive indexes of the second reflective member R2 and the third reflective member R3 may be indicated by negative numbers, which may indicate that the second reflective

member R2 and the third reflective member R3 may have a reflective surface for reflecting light.
(270) TABLE-US-00017 TABLE 17 Radius Thickness Surface of or Refractive Abbe No. Note curvature distance index number S1 First Infinity 2.4500 1.7 29 reflective member S2 Infinity 2.4500 1.7 29 S3 Infinity 1.9420 S4 First -6.8132 1.0000 1.671 20 S5 lens -5.9584 4.2926 S6 Second -5.2687 1.0000 1.544 56 S7 lens 19.2779 0.1384 S8 Stop Infinity 0.1424 S9 Third -23.0600 1.4517 1.544 56 S10 lens -5.3270 0.1565 S11 Fourth -8.8103 1.0000 1.595 31.1 S12 lens 12.2400 0.3974 S13 Fifth 9.4930 1.9278 1.544 56 S14 lens -6.3940 D56 S15 Sixth R11 T6 1.298 100 S16 lens Infinity 0.5000 1.526 55 S17 Infinity D6R S18 Second Infinity 0.0000 -1 reflective member S19 Infinity -7.0000 S20 Third Infinity 0.0000 -1 reflective member S21 Infinity DR3F S22 Filter Infinity 0.2100 1.519 64.2 S23 Infinity 1.9631 S24 Imaging Infinity 0.0000 surface

(271) TABLE-US-00018 TABLE 18 In case of first In case of second focal length focal length R11 18.5 -42.86 D56 1.7982 2.2484 T6 1.8151 1.3649 D6R 3 13.1809 DR3F 7.5469 17.7278

(272) R11 is the radius of curvature of the object-side surface of the sixth lens **560**, D56 is the distance on the optical axis from the image-side surface of the fifth lens **550** to the object-side surface of the sixth lens **560**, T6 is the thickness on the optical axis of the sixth lens **560**, D6R is the distance on the optical axis from the image-side surface of the sixth lens **560** to the second reflective member R2, and DR3F is distance on the optical axis between the third reflective member R3 and the filter **570**.

(273) TABLE-US-00019 TABLE 19 In case of first In case of second focal length focal length f 16.1726 31.755 f1 48.216963 48.216963 f2 -7.503543 -7.503543 f3 12.38179 12.38179 f4 -8.47084 -8.47084 f5 7.339655 7.339655 f6 62.10051479 -143.871787

(274) The definitions of f, f1, f2, f3, f4, f5, and f6 may be the same as those of the first example.

(275) In the fifth example, the first lens **510** may have positive refractive power, the first surface of the first lens **510** may be concave, and the second surface of the first lens **510** may be convex.

(276) The second lens **520** may have negative refractive power, and the first and second surfaces of the second lens **520** may be concave.

(277) The third lens **530** may have positive refractive power, the first surface of the third lens **530** may be concave, and the second surface of the third lens **530** may be convex.

(278) The fourth lens **540** may have negative refractive power, and the first and second surfaces of the fourth lens **540** may be concave.

(279) The fifth lens **550** may have positive refractive power, and the first and second surfaces of the fifth lens **550** may be convex.

(280) The sixth lens **560** may be a variable lens of which a focal length may change. For example, the sixth lens **560** may have a first focal length and a second focal length. The sixth lens **560** may include a liquid lens **561** and a flat lens **563** attached to an image-side surface of the liquid lens **561**.

(281) When the sixth lens **560** has the first focal length, the sixth lens **560** may have positive refractive power. Also, the first surface of the sixth lens **560** (e.g., the object-side surface of the liquid lens **561**) may be convex, and the second surface of the sixth lens **560** (e.g., the image-side surface of the flat lens **563**) may be flat.

(282) When the sixth lens **560** has the second focal length, the sixth lens **560** may have negative refractive power. Also, the first surface of the sixth lens **560** may be concave, and the second surface of the sixth lens **560** may be flat.

(283) Each surface of the first lens **510** to the fifth lens **550** may have an aspherical coefficient as in Table 20. For example, the object-side surfaces and the image-side surfaces of the first to fifth lenses **510** to **550** may be aspherical.

(284) TABLE-US-00020 TABLE 20 Conic Fourth Sixth Eighth Tenth Twelfth constant coefficient coefficient coefficient coefficient (K) (A) (B) (C) (D) (E) S4 0 0.000635779 -3.37E-05 -3.03E-06 -1.12E-06 1.10E-07 S5 0 0.000954221 -1.28E-06 -9.04E-06 -1.29E-07 4.93E-08

S6 0 0.000729294 6.23E-06 4.76E-06 -3.62E-06 4.47E-07 S7 0 -2.54065E-05 -1.60E-05
7.30E-07 7.67E-07 2.38E-07 S9 0 -0.000446111 -1.80E-05 -2.82E-06 -9.16E-07 3.34E-08
S10 0 0.000274266 -1.60E-05 -4.55E-07 -4.24E-07 -1.59E-07 S11 0 -3.61E-04
-1.95E-05 -2.95E-06 -6.71E-07 8.96E-08 S12 0 -7.11911E-06 -2.15E-05 -5.05E-06
8.04E-08 -7.97E-09 S13 0 -1.57883E-05 -3.65E-06 -1.12E-06 5.44E-08 3.74E-09 S14 0
8.26479E-05 2.76E-05 -1.17E-06 2.36E-07 4.49E-09

(285) Also, the optical imaging system configured as described above may have aberration properties illustrated in FIGS. **18** and **20**.

(286) According to the aforementioned examples, the optical system may implement a zoom function by changing the focal length.

(287) While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

Claims

1. An optical imaging system, comprising: a plurality of fixed lenses disposed along an optical axis; a first reflective member disposed on an object side of the plurality of lenses; and a plurality of reflective members disposed on an image side of the plurality of lenses, wherein at least one of the plurality of lenses is a variable lens having a variable focal length, wherein each of the plurality of reflective members disposed on the image side of the plurality of lenses is configured to move as the focal length of the variable lens changes, wherein $-7 < (fv_1 \times D6R_2) / (fv_2 \times D6R_1) < -1$, where fv_1 is a first focal length of the variable lens, fv_2 is a second focal length of the variable lens, $D6R_1$ is a distance on an optical axis from a rearmost lens of the plurality of lenses to a reflective member most adjacent to the rearmost lens among the plurality of reflective members when the variable lens has the first focal length, and $D6R_2$ is a distance on an optical axis from the rearmost lens to the reflective member most adjacent to the rearmost lens when the variable lens has the second focal length, wherein the plurality of lenses includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, and a sixth lens disposed in order from the object side, wherein the first to sixth lenses are spaced apart from each other along the optical axis, wherein the optical imaging system has a total of six lenses, and wherein the first lens has positive refractive power and a concave object-side surface in a paraxial region thereof.
2. The optical imaging system of claim 1, wherein the variable lens is configured to have a first focal length or a second focal length as the focal length changes, and wherein the first focal length is positive and the second focal length is negative.
3. The optical imaging system of claim 1, wherein the variable lens comprises a liquid lens and a flat lens attached to an image-side surface of the liquid lens.
4. The optical imaging system of claim 3, wherein a radius of curvature of an object-side surface of the liquid lens is variable and a thickness on an optical axis of the liquid lens is variable.
5. The optical imaging system of claim 4, wherein the variable lens is configured to have a first focal length or a second focal length as the focal length changes, the first focal length is positive, and the second focal length is negative, wherein, when the variable lens has the first focal length,

the radius of curvature of the object-side surface of the liquid lens is positive, wherein, when the variable lens has the second focal length, the radius of curvature of the object-side surface of the liquid lens is negative, and wherein the thickness on the optical axis of the liquid lens is smaller when the variable lens has the second focal length than when the variable lens has the first focal length.

6. The optical imaging system of claim 1, wherein the variable lens is the second lens or the sixth lens.

7. The optical imaging system of claim 1, wherein the variable lens is the second lens, and wherein the second lens has positive or negative refractive power, the third lens has negative refractive power, the fourth lens has positive refractive power, the fifth lens has negative refractive power, and the sixth lens has positive refractive power.

8. The optical imaging system of claim 1, wherein the variable lens is the sixth lens, and wherein the second lens has negative refractive power, the third lens has positive refractive power, the fourth lens has negative refractive power, the fifth lens has positive refractive power, and the sixth lens has positive or negative refractive power.

9. The optical imaging system of claim 1, wherein the plurality of reflective members includes a second reflective member and a third reflective member, and wherein the second reflective member comprises a reflective surface configured to reflect light in a direction perpendicular to an optical axis of the plurality of lenses, and the third reflective member comprises a reflective surface configured to reflect light reflected by the second reflective member in a direction parallel to the optical axis of the plurality of lenses.

10. The optical imaging system of claim 9, wherein the second reflective member and the third reflective member are configured to move together.

11. The optical imaging system of claim 1, wherein $10 < f_{v_1}/D_{6R_1}$.

12. The optical imaging system of claim 1, wherein $-15 < f_{v_2}/D_{6R_2} < -3$.

13. The optical imaging system of claim 1, wherein $-3 < f_{v_2}/f_{v_1} < 0$.

14. The optical imaging system of claim 1, wherein $3 < D_{6R_2}/D_{6R_1} < 6$, and wherein the first focal length is positive and the second focal length is negative.

15. The optical imaging system of claim 1, wherein $0 < L1/TTL1 < 1$, where $L1$ is a linear distance in a direction parallel to an optical axis from the object-side surface of the first lens to an imaging surface when the variable lens has a first focal length, and $TTL1$ is a distance on the optical axis from the object-side surface of the first lens to the imaging surface when the variable lens has the first focal length.

16. The optical imaging system of claim 1, wherein $0 < L2/TTL2 < 1$, where $L1$ is a linear distance in a direction parallel to an optical axis from the object-side surface of the first lens to an imaging surface when the variable lens has a second focal length, and $TTL2$ is a distance on the optical axis from the object-side surface of the first lens to the imaging surface when the variable lens has the second focal length.

17. The optical imaging system of claim 1, wherein $1 < (L1 \times TTL2)/(L2 \times TTL1) < 3$, where $L1$ is a linear distance in a direction parallel to an optical axis from the object-side surface of the first lens to an imaging surface when the variable lens has a first focal length, $L2$ is a linear distance in the direction parallel to the optical axis from the object-side surface of the first lens to the imaging surface when the variable lens has a second focal length, $TTL1$ is a distance on the optical axis from the object-side surface of the first lens to the imaging surface when the variable lens has the first focal length, and $TTL2$ is a distance on the optical axis from the object-side surface of the first lens to the imaging surface when the variable lens has the second focal length.
